Western Music Theory Fundamentals: Study Notes

Subject: Pitch, Staves, Clefs, Diatonic Scales, and Intervals | *Educational Reference Guide*

Section 1: Pitch, Staff Notation, and Clefs

Musical sounds are graphically recorded on a five-line staff. The pitch of a tone indicates its frequency of vibration.

Clefs:
• **Treble Clef (G Clef):** Loops around the second line, establishing it as G4 (392 Hz), immediately above middle C.
• **Bass Clef (F Clef):** Two dots surround the fourth line, fixing it as F3 (174.6 Hz), directly below middle C.
• **Alto Clef (C Clef):** The center indentation defines middle C (C4) on the third line (used primarily for viola).
• **Tenor Clef (C Clef):** Indentation places middle C (C4) on the fourth line (used for cello, trombone, bassoon in high registers).

Scientific Pitch Notation: Middle C is designated C4. Octaves run from C to B. The octave below middle C is octave 3 (C3 to B3); the octave above is octave 5 (C5 to B5).

Section 2: Diatonic Major and Minor Scales

A diatonic scale consists of seven distinct pitch classes arranged in alphabetical order within an octave.

The Major Scale Formula:
The major scale follows the invariant sequence: **Tone – Tone – Semitone – Tone – Tone – Tone – Semitone** (W – W – H – W – W – W – H).
Semitones occur naturally between scale degrees 3–4 and 7–8. Scale degree names are: 1 Tonic, 2 Supertonic, 3 Mediant, 4 Subdominant, 5 Dominant, 6 Submediant, 7 Leading Tone.

Minor Scale Forms:
• **Natural Minor:** Step pattern W – H – W – W – H – W – W (1 – 2 – b3 – 4 – 5 – b6 – b7).
• **Harmonic Minor:** The 7th degree is raised by one semitone both ascending and descending (1 – 2 – b3 – 4 – 5 – b6 – 7). This creates an augmented second (3 semitones) between degrees 6 and 7, establishing a half-step leading tone resolution to the tonic.
• **Melodic Minor:** Ascending, both 6th and 7th degrees are raised (1 – 2 – b3 – 4 – 5 – 6 – 7); descending, it reverts to the natural minor scale.

Section 3: Intervals and Semitone Calculations

An interval measures the distance between two pitches reckoned upwards from the lower note. Intervals are classified by numerical size and quality (Perfect, Major, Minor, Augmented, Diminished).

Table of Standard Intervals by Semitones:

• Perfect Unison: 0 semitones
• Minor Second: 1 semitone (e.g. C to Db)
• Major Second: 2 semitones (e.g. C to D)
• Minor Third: 3 semitones (e.g. C to Eb)
• Major Third: 4 semitones (e.g. C to E)
• Perfect Fourth: 5 semitones (e.g. C to F)
• Augmented Fourth / Diminished Fifth (Tritone): 6 semitones (e.g. C to F# is Augmented 4th; C to Gb is Diminished 5th)
• Perfect Fifth: 7 semitones (e.g. C to G)
• Minor Sixth: 8 semitones (e.g. C to Ab)
• Major Sixth: 9 semitones (e.g. C to A)
• Minor Seventh: 10 semitones (e.g. C to Bb)
• Major Seventh: 11 semitones (e.g. C to B)
• Perfect Octave: 12 semitones (e.g. C to C)